Why Do Some Models Perform Better?

A Comparative Study of Logistic Regression, Random Forest and XGBoost for Classification

MODULE: MACHINE LEARNING AND NEURAL NETWORKS

GitHub: https://github.com/gurralalaxminarasimha/student-performance-classifier

ABSTRACT

Choosing the appropriate machine learning model is one of the basic concerns in data science applications. Various machine learning models have different presumptions about data and different levels of interpretability, robustness, and sensitivity to hyperparameters. This report is focused on exploring how different classification algorithms, including Logistic Regression, Random Forest, and XGBoost, perform differently on the same data and why some algorithms perform better than others.

The UCI Student Performance dataset is used in the study. The dataset has social and behavioral characteristics of secondary school students. Using the dataset, the study predicts if a student will pass or fail in the final examination. Various visualizations and evaluations are done to compare the models based on different parameters.

The results show that none of the models performs better than the others across all evaluation criteria. The highest AUC score is obtained by the Logistic Regression model, which is 0.841, making it the best model for ranking students by risk level. The Random Forest model offers the highest accuracy with the least tuning required. The XGBoost model, though strong, should be used with caution to avoid suboptimal results. The results show that the choice of the model should be task-dependent, i.e., depending on whether the results should be interpreted, the accuracy of the results, or the stability of the results is of concern.

INTRODUCTION

In various fields such as healthcare, finance, education, and social sciences, machine learning algorithms have been successfully used to analyze data and make decisions. Classification is one type of machine learning algorithm that tries to classify data into classes based on features. Out of all the classification algorithms, three prominent ones are Logistic Regression, Random Forest, and XGBoost.

Each of the above models employs a distinct method for learning the classification boundary. The LR model uses a linear boundary to classify the data, which makes the model transparent and interpretable. The RF model uses multiple decision trees, each of which is trained on a random subset of the data, to achieve diversity and hence robustness. The XGBoost model uses a sequence of decision trees, each of which refines the results of the previous tree to correct its errors, to achieve high accuracy (Chen and Guestrin, 2016).

However, it has been found that practitioners tend to choose the model based on convention, not on any specific understanding of the reasons for the performance difference between the models. The present report aims to bridge the gap by comparing all three models on the same data set and understanding the reasons for the performance difference.

Three models were examined in this study:

1. **Logistic Regression:** Logistic Regression fits a linear boundary to the log-odds of the outcome. It is computationally efficient and highly interpretable but may underperform when the true decision boundary is non-linear.
2. **Random Forest:** Random Forest builds an ensemble of decision trees, each trained on a random bootstrap sample and a random subset of features. The final prediction is a majority vote, which reduces overfitting and handles non-linear patterns well (Breiman, 2001).
3. **XGBoost:** XGBoost constructs trees sequentially in a gradient-boosting framework. Each tree learns to correct the residual errors of the previous model, resulting in a highly accurate but parameter-sensitive classifier (Chen and Guestrin, 2016).

BACKGROUND OF THE THREE MODELS

The three models used in this study belong to different classes of machine learning algorithms, each based on different theory.

Logistic regression is one of the oldest and best-understood classification algorithms. Initially designed for binary outcomes, logistic regression is used to predict the probability that a case belongs to a particular class based on a sigmoid function of a linear combination of features (Hosmer, Lemeshow, and Sturdivant, 2013). Though one of the simplest methods, it is highly competitive when linear relationships between features and outcomes can be assumed and when it is important to know the exact effect of each feature.

Random Forest is an ensemble algorithm developed by Breiman (2001), which is an extension of decision trees and uses the technique of bagging and feature Random Forests. Bagging is the technique of using different subsets of the data to train different decision trees. This helps in reducing the variance of the decision trees, and the algorithm works well in the context of generalisation. In Random Forests, the feature importance of each feature is calculated on the basis of the reduction in impurities.

XGBoost is the implementation of the gradient boosting algorithm developed by Chen and Guestrin (2016). Unlike the Random Forest algorithm, XGBoost works on the technique of boosting, in which the trees are fitted one by one on the error of the previous ensemble of trees. This makes the algorithm suitable for structured data, but the parameters need to be adjusted properly in the context of the algorithm to prevent overfitting.

The current machine learning best practice is that there is no single algorithm that is best for all tasks; rather, the best algorithm depends on the dataset, the evaluation metric, and the interpretability needs (James et al., 2023). This study provides empirical support for this best practice using a real social dataset.

DATASET AND EXPERIMENTAL SETUP

In this paper, the UCI Student Performance dataset (Cortez and Silva, 2008), which contains information on 649 secondary school students, has been used. The data set has information on various social, behavioral, and family background attributes collected through surveys. The target variable is based on the final examination grade (G3) of the student, where the student is considered Pass (1) if the grade is 10 or higher, and Fail (0) if the grade is lower than 10.

The dataset was chosen for the analysis because it represents a realistic problem for binary classification, with detailed behavioral information such as the amount of time studying, the amount of alcohol consumed, the frequency of going out, family relationships, and parental education, all of which are relevant for the evaluation of the interpretability of the models and the feature importance.

The analysis was implemented using Python and several scientific computing libraries. The main tools used in this project include:

- Pandas and NumPy for data loading and numerical operations
- Matplotlib for all visualisations, using a colourblind-friendly palette (Wong, 2011)
- Scikit-learn for Logistic Regression, Random Forest, and evaluation metrics (Scikit-learn Developers, 2024)
- XGBoost library for the gradient-boosted classifier (Chen and Guestrin, 2016)

The dataset was split into 70% for training and 30% for testing, and stratified sampling was used to preserve class balance in the data. The categorical variables, namely school, sex, and address, were encoded using label encoding before the model was trained. The raw grade column, i.e., G3, was excluded from the feature list to avoid data leakage.

RESULTS AND VISUALISATION

1. Dataset Distribution

The distribution of the dataset and the major behavioural characteristics can be seen in Figure 1 below. The class distribution shows a small degree of bias, with 362 students passing and 287 students failing the exam. The study time bar chart shows a strong positive correlation between the amount of study time and the percentage of students who passed the exam. The weekend alcohol consumption is right-skewed, indicating that the majority of the students consumed very little alcohol.

Figure 1: Student Dataset Overview

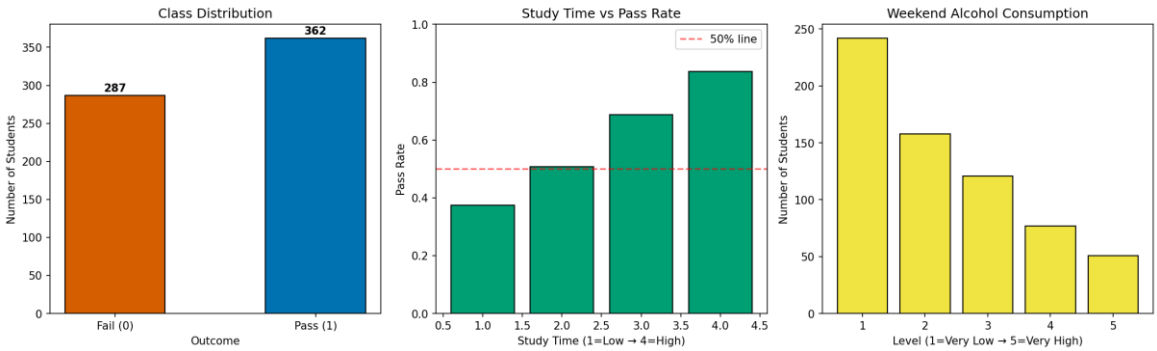

Figure 1: Overview of the student dataset showing class distribution, pass rate by study time, and weekend alcohol consumption levels.

2. Exploratory Data Analysis

Figure 2 shows a detailed comparison of six important behavioural features between students who have passed and students who have failed. From the analysis, several important trends are identified.

More failed students are likely to fail if they have had more past failures. Similarly, students who have studied more consistently are likely to pass. More weekend alcohol usage and going out are associated with students who have failed. Better family relationship quality has a slight positive association with passing. This shows that the data set has meaningful and well-structured relationships.

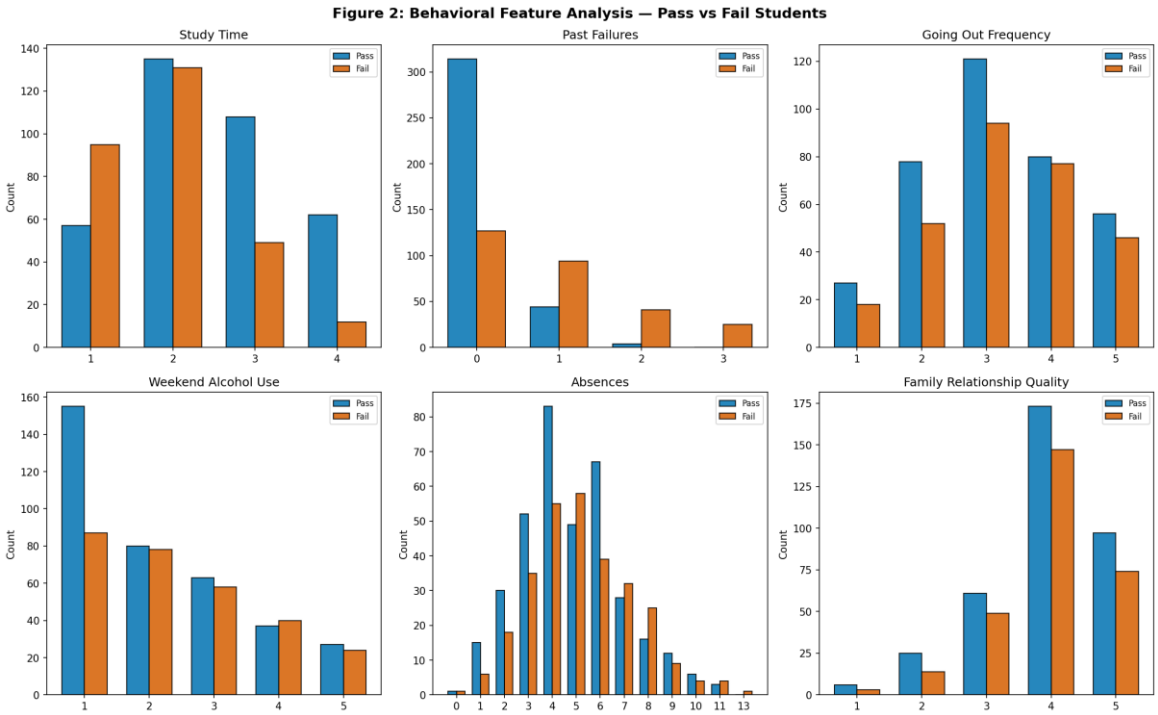

Figure 2: Comparison of six behavioural features between students who passed and failed. Key predictors include past failures, study time, and alcohol use.

3. Model Training and Accuracy Comparison

The three models have been trained using the same training set and have been tested using the held-out set. Figure 3 shows the accuracy of each model for classification. From the results obtained in the figure, it can be seen that the accuracy for all three models is quite similar. The accuracy for each model varies between 0.74 and 0.76. The highest accuracy is obtained by Random Forest (0.759), followed by Logistic Regression (0.754), and the lowest accuracy is obtained by XGBoost (0.738).

This similarity in the accuracy of the three models implies that the dataset does not have a problem that can be solved by one of the algorithms more than the others. However, as seen in the subsequent sections, the three models vary significantly in other aspects.

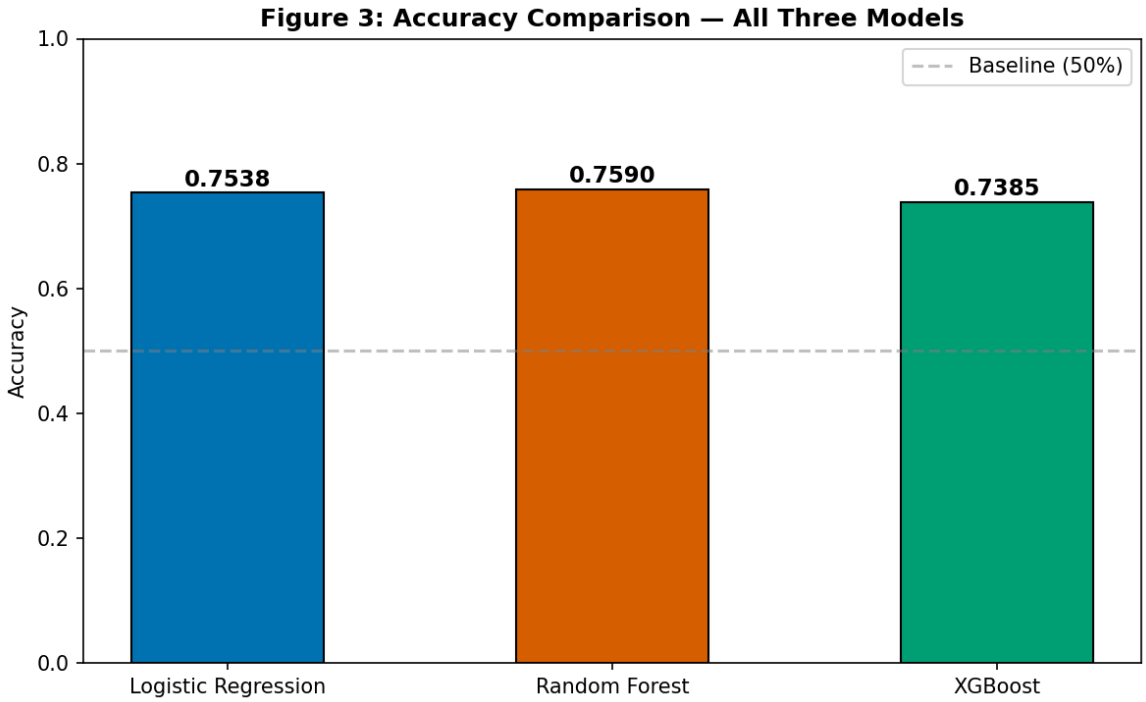

Figure 3: Bar chart comparing classification accuracy across Logistic Regression, Random Forest, and XGBoost on the test set.

4. Confusion Matrix Analysis

The confusion matrices in Figure 4 give further information about each of these models' classification behavior. All three models appear to favor classification of the majority class (Pass) more often, as indicated by the small imbalance in the dataset. The false negative rate of failing students who were predicted as passing is consistent among all three models. This indicates that the imbalance of the dataset is affecting these results more than the algorithm used.

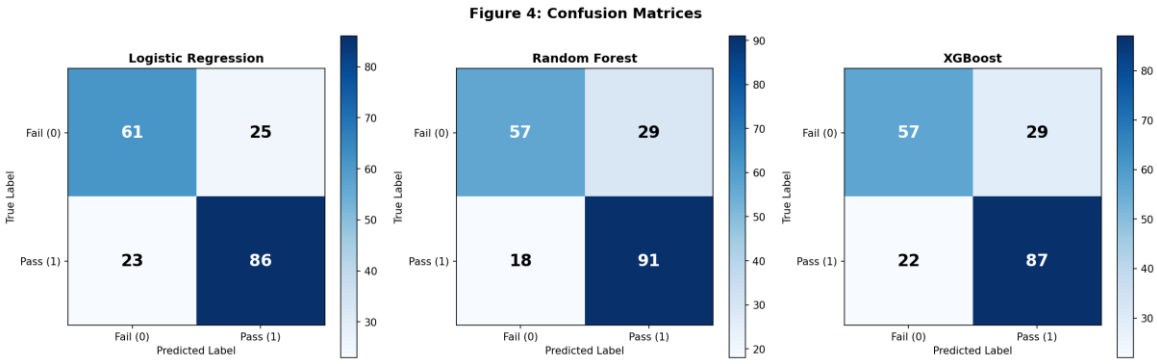

Figure 4: Confusion matrices for all three models. The diagonal cells show correct predictions; off-diagonal cells show misclassifications.

5. ROC Curve Comparison

Figure 5 shows the ROC curves for all three models. The AUC score is a threshold-independent score for the quality of each model's ranking of the students by the probability of passing. It turns out that the highest AUC score is for the Logistic Regression model (0.841), followed by the Random Forest (0.821), and then the XGBoost (0.801) model.

This is interesting, as it shows that the simplest model, the Logistic Regression, is the best at discriminating between the passing and failing students at all possible thresholds. This is consistent with the nearly linear relationship between the behavioural features and the outcome, as seen in the EDA above. In fact, as the relationship between the inputs and the outcome is already nearly linear, the more complex models do not seem to help.

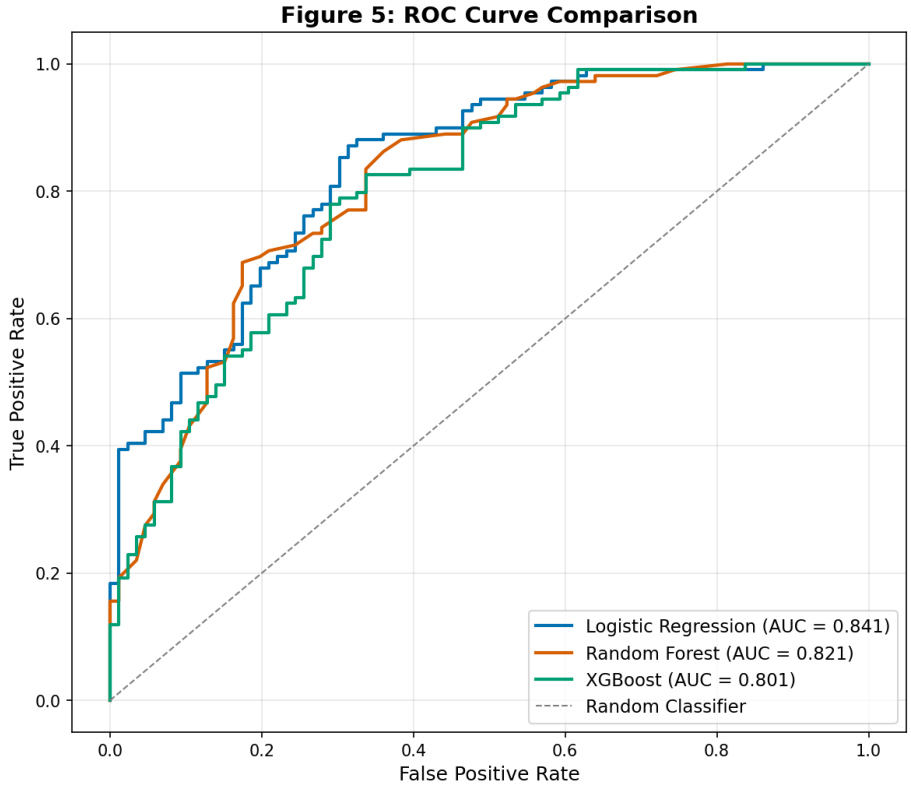

Figure 5: ROC curves for all three models. Logistic Regression achieves the highest AUC (0.841), demonstrating the best overall discrimination ability.

6. Feature Importance

Figure 6 displays the feature importance scores given by Random Forest and XGBoost. Both tree-based classifiers are in agreement that past failures/failures and absence are the most important factors. However, it is noticeable that XGBoost gives slightly more weight to alcohol consumption-related features Walc and Dalc compared to Random Forest.

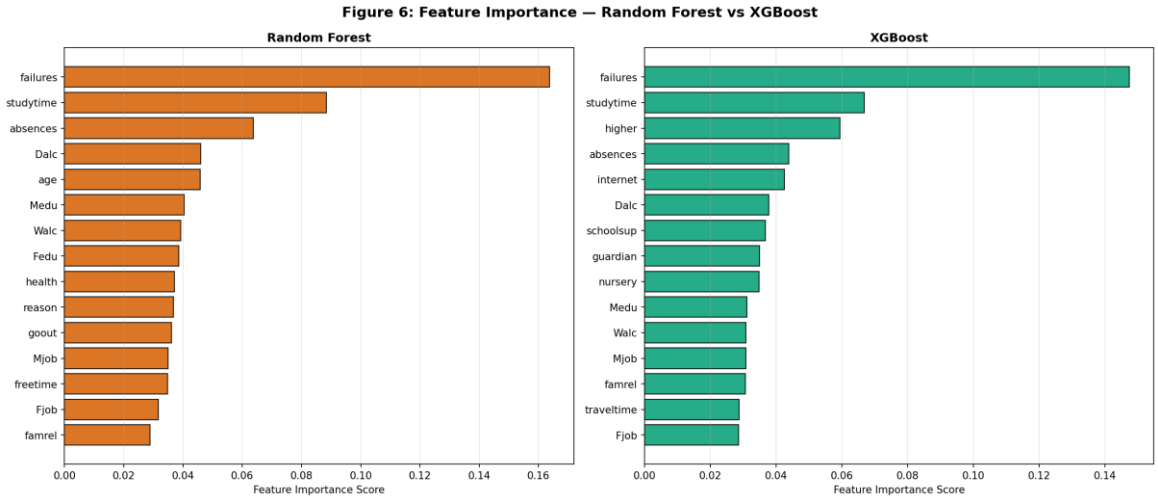

Figure 6: Feature importance scores for Random Forest and XGBoost. Both models identify past failures and absences as the most important predictors.

In Figure 7, we can see the coefficients of the features as calculated by the Logistic Regression algorithm. Note that, unlike the other models, the coefficients have directional implications: the features with negative coefficients have a decreasing effect on the probability of passing (notably failures, Walc, and goout), whereas the features with positive coefficients have increasing effects on the probability of passing (notably studytime and higher – the student's aspiration for higher education).

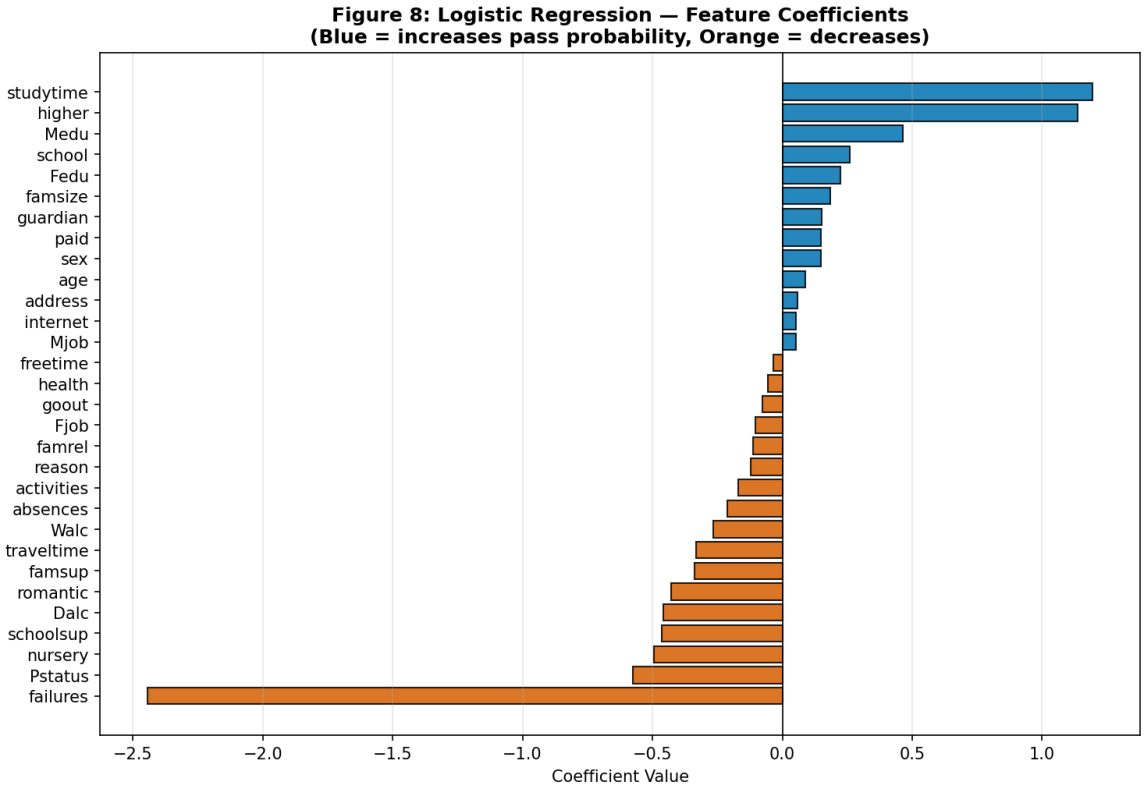

Figure 7: Logistic Regression feature coefficients. Blue bars increase pass probability; orange bars decrease it. Past failures and alcohol use are the strongest negative predictors.

7. Hyperparameter Tuning

Figure 8 depicts the sensitivity of the performance of the Random Forest and XGBoost classifiers to their respective key parameters. As depicted, the performance of the Random Forest classifier increases with the number of estimators (the number of decision trees), but only up to 50. Increasing the number of estimators beyond 50 has no significant effect on the performance. Thus, the Random Forest classifier is not very difficult to tune.

The XGBoost classifier, on the contrary, is very sensitive to the learning rate. As depicted, the learning rate should not be too high (0.5), and the optimal performance is obtained when the learning rate is around 0.1.

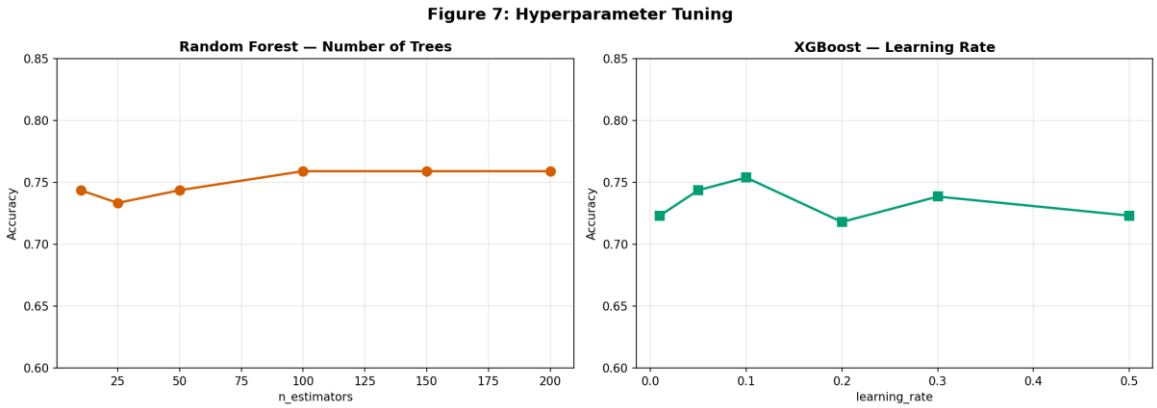

Figure 8: Hyperparameter sensitivity analysis. Random Forest stabilises quickly with more trees, while XGBoost is more sensitive to learning rate selection.

DISCUSSION

The results of this study demonstrate that model selection involves a trade-off between multiple competing criteria, and that choosing a model based on accuracy alone can be misleading.

The similarity in classification accuracy across all three models (0.74 to 0.76) suggests that the dataset's structure is relatively straightforward and does not require the complexity of ensemble methods or sequential boosting. This is consistent with the EDA findings, which showed clear and approximately monotonic relationships between behavioural features and pass probability. In such cases, Logistic Regression — with its linear decision boundary — is often sufficient.

The AUC comparison reinforces this interpretation. Logistic Regression's superior AUC score (0.841) indicates that it provides the most reliable ranking of students by risk, which is the most practically relevant metric in an educational intervention context where identifying at-risk students is the primary goal.

The feature importance analysis reveals a high degree of agreement between the three models about which features matter most. Past failures, absences, and study time emerge as the top predictors across all three approaches. This convergence across different algorithmic families increases confidence that these are genuinely influential behavioural factors, rather than artefacts of any particular model's assumptions.

The hyperparameter analysis highlights an important practical consideration: XGBoost's sensitivity to learning rate means that using default settings without tuning can lead to worse performance than a simpler, less parameter-sensitive model. Practitioners should account for the computational cost of tuning when deciding which model to use.

Summary of Model Comparison

Table 1: Summary comparison of all three models across key evaluation dimensions.

Model	Accuracy	AUC Score	Interpretability	Tuning Required
Logistic Regression	0.7538	0.8412	High (coefficients)	Low
Random Forest	0.7590	0.8206	Medium (importance)	Low-Medium
XGBoost	0.7385	0.8013	Medium (importance)	High

CONCLUSION

This report aimed to examine the reasons behind the varying results obtained by different machine learning models when applied to the same dataset. In this case, the dataset was behavioral and social student data that was used to predict pass or fail results. Three models were used in the report. They included Logistic Regression, Random Forest, and XGBoost. The results obtained from the experiments showed that the choice of model depends on some compromises that go beyond accuracy. It was evident that while Logistic Regression was the best in terms of ranking students by risk (highest AUC), it was also the most interpretable (directional coefficients) and needed the least tuning. Random Forest was the best in terms of accuracy while needing minimal configuration. It also produced reasonable feature importance values. In addition, XGBoost was also a great model but needed tuning and produced no

accuracy gains. It was evident that across all three models, past failures, absences, and study time were consistently ranked as the most important features for predicting student performance. The main practical takeaway from this work is that the choice of model should be informed by the needs of the task. If interpretability is important – as in educational settings, medicine, and policy – then Logistic Regression is a great choice. If robustness to non-linearities is important, Random Forest is a safe bet. XGBoost should be reserved for when you need the highest accuracy and have the computational power to fine-tune.

ACCESSIBILITY AND ETHICAL CONSIDERATIONS

All visualisations in this research have been created using a colorblind-friendly color scheme inspired by Wong (2011), which used colors #0072B2 (blue), #D55E00 (orange), and #009E73 (green). Axis labels, titles, and legends have been added to all visualisations to help users with visual impairments.
Ethics of predicting student performance:
The ethical implications of predicting student performance have to be considered for the application of this model:
• Reinforcing Disadvantage: Features such as parental education level, family status, and location of residence may be indirect indicators of socioeconomic status.
• Labeling Effects: Informed students that they will likely perform poorly may influence the teacher or school's attitude towards them, which could result in a self-fulfilling prophecy.
• Data Consent and Privacy: The data of the students is sensitive information, and consent from the students is required for the application of this predictive model, which needs to comply with UK GDPR regulations.
• Purpose of Prediction: The purpose of a model like this should be to facilitate intervention and resource allocation, not to classify and marginalize students. Models like this, which are easily interpretable like Logistic Regression, are well suited to this purpose, especially when their logic can be interrogated and challenged by educators themselves.

These points are all relevant to the EU's classification of AI systems used in education as high-risk applications that need to be audited for bias (European Commission, 2021).

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